

MUIZ ABIOLA AMINU

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RESEARCH INTERESTS

Control Systems, Embedded Architectures, Autonomous Mobile Robotics, Edge Intelligence (TinyML), and Mechatronic Integration for Industrial, Renewable Energy Systems.

EDUCATION

Federal University of Technology, Minna

Minna, Nigeria

Bachelor of Engineering (B.Eng.) in Mechatronics Engineering

Dec 2025

Second Class Honours (Upper Division) | **CGPA:** 4.37 / 5.00 (87%)

Capstone Thesis: *Development of a Semi-Autonomous Navigation System for a Pipeline Monitoring Robot*

PUBLICATIONS & CONFERENCE PRESENTATIONS

- Aminu, M. A.** (2026). "Development of a Semi-Autonomous Navigation System for a Pipeline Monitoring Robot." *Preprint*. [\[Preprint PDF\]](#)
- Aminu, M. A.** (2026). "Hardware-in-the-Loop Benchmarking of Edge-NILM on RISC-V Silicon: Decentralized Demand-Side Management for Resilient African Solar Mini-Grids." *Accepted for Poster Presentation, BUILD26 Conference*, Abuja, Nigeria. [\[Paper PDF\]](#)

RESEARCH & ENGINEERING EXPERIENCE

EN1 Technologies | *Embedded Systems Engineer*

Jun 2026 – Present

- Architect bare-metal STM32F407 firmware for e-tricycle ECUs and develop ESP32 thermoforming control systems integrating SSR loops and MQTT telemetry.

Aflon Center for Artificial Intelligence and Robotics | *IoT Research Engineer*

Jan 2026 – Jun 2026

- Prototyped IoT hardware and designed architectures bridging microcontroller telemetry with cloud platforms.

Advanced Engineering Innovation Research Group | *Research Assistant*

Jul 2021 – Dec 2025

- Evaluated experimental setups for autonomous obstacle detection, sensor fusion, and wireless telemetry links.

Ideal Intelligent Innovation Research Group | *Research Associate*

May 2022 – Dec 2025

- Prototyped hardware and optimized motor control logic for automation projects while mentoring student researchers.

Artificial Intelligence for Clean Energy Ltd. | *Engineering Intern*

Jun 2024 – Nov 2024

- Engineered smart energy telemetry hardware for clean energy monitoring and microgrid applications.

E-lord Technology | *Hardware Engineer (Part-Time)*

Aug 2019 – Present

- Execute schematic capture, footprint assignment, and multi-layer PCB layout utilizing EasyEDA for prototypes.

SELECTED ENGINEERING PROJECTS

- Semi-Autonomous Pipeline Inspection Robot:** Engineered an inspection rover using Raspberry Pi 4, featuring ultrasonic hazard detection (1.02s halt response) and bidirectional RESTful API teleoperation via React/Node.js.
- Edge AI Microcontroller Benchmarking:** Benchmarked TensorFlow Lite for Microcontrollers (TinyML) inference latency, memory footprint, and int8 quantization performance across ESP32-C3 architectures.
- Life-Guard Node (Symbiotic Pipeline Security):** Engineered a solar-powered telemedical proof-of-concept integrating Gemini AI for predictive health diagnostics of the pipeline host communities. Designed a LoRa safety interlock that autonomously severs power and dispatches alerts during simulated pipeline breaches.

TECHNICAL SKILLS

- Embedded Systems & SBCs:** ARM Cortex-M (STM32F407), ESP32, AVR, Raspberry Pi.
- Firmware & Programming:** Bare-metal C/C++, CMSIS, FreeRTOS, Python, Dart (Flutter), MATLAB/Simulink.
- Hardware Design:** Multi-layer PCB Layout (KiCad, EasyEDA), Stepper/Servo Drivers, SSRs, Power Electronics.
- IoT Protocols:** MQTT (Mosquitto), REST APIs, Firebase, Node-RED, UART, SPI, I2C.

CERTIFICATIONS & AFFILIATIONS

- Memberships:** Professional Member, Society of Petroleum Engineers (SPE); Graduate Member, Nigerian Society of Engineers (GMNSE).
- Certifications:** Simulink Fundamentals (MathWorks); Career Essentials in Sustainable Tech (Microsoft).
- Leadership:** Lead STEM Facilitator (Build-a-thon, 2024–2025) – instructed 250+ students in embedded hardware.